Leo Orshansky

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Pursuing a doctorate degree in theoretical computer science.

Education

2024 – Present **Ph.D. in Theoretical Computer Science, Columbia University**

2024 – 2026 **M.S. Computer Science, Columbia University**

2020 – 2024 **B.S. Computer Science (Turing Scholars Honors), University of Texas at Austin**
Honors thesis title: *Computation in Isolated Thermodynamic Systems*.
B.S. Mathematics, University of Texas at Austin
Minor in Chinese Language, University of Texas at Austin

Research Experience

2024 – Present **PhD Research in Quantum Cryptography,** Columbia University

- *Doctoral Research, Advised by Profs. Tal Malkin and Henry Yuen:* Collaborating with researchers at Columbia, along with other institutions, to advance theoretical knowledge of quantum position verification, information-theoretic cryptography, secure multiparty computation, and other topics in the intersection of quantum computation and cryptography.

2022 – 2024 **Undergraduate Computer Science Research,** University of Texas at Austin

- *Undergraduate Honors Thesis, Supervised by Prof. David Soloveichik (Fall 2023):* Explored the connection between computational power of unconventional computing models (such as biochemical and reversible computing) and thermodynamic isolation. Co-wrote a research paper on a new framework for zero-energy computational sampling using the model of Brownian computation.

Employment Experience

2026 – 2027 **Quantum Research Intern,** Hon Hai Research Institute (Foxconn). Remote
Researching various theoretical directions, primarily within the foundations of quantum cryptography (microcrypt). Supervised by Min-Hsiu Hsieh and collaborating with other members of the team, including Taiga Hiroka.

2024, Jun – Aug **Software Engineering Intern,** Citadel Securities. New York, NY
Created a fully customizable, performant, and extensible alerting platform for options traders, built atop the highly performant KDB+ timeseries database and Q query language. Rolled out several high-priority alerts to traders, alongside a customization UI.

2023, Jun – Aug **Software Engineering Intern,** Jane Street Capital. New York, NY
Added sidecar metrics such as per-symbol risk usage and tracking of inflight messages to the firm's primary order engine in OCaml and enabled publishing them to Prometheus + Grafana. Developed a new OCaml tool which enables root-privileged OCaml unit tests by transparently sandboxing them inside of QEMU virtual machines, and integrated this with the build system.

Employment Experience (continued)

2022, Jun – Aug
- **Software Engineering Intern,** Bloomberg LP. New York, NY
  Rebuilt the "Mortgages Architecture Dispatcher" in Rust — a dynamically configurable reverse proxy providing transparent caching, UUID-based routing, and API versioning to the mortgages service mesh. Results: 99% reduction in dispatcher memory usage, and 50% speedup in cache hit latency.

2021, Jun – Aug
- **Software Engineering Intern,** Cloudflare, Inc. Austin, TX
  Designed and implemented an extension set for the Cloudflare Workers (serverless JS/WASM platform) Rust bindings. Deployed a Kubernetes application to host a new automated submission + grading system for the Cloudflare hiring assignment.

Research Publications

1. U. Girish, A. May, **L. Orshansky**, and C. Waddell, "Comparing classical and quantum conditional disclosure of secrets," *Quantum*, vol. 10, p. 2049, Apr. 2026, ISSN: 2521-327X. DOI: `10.22331/q-2026-04-01-2049`.
2. U. Girish, S. Goldwasser, G. Gluch, T. Malkin, **L. Orshansky**, and H. Yuen, "Private proofs of when and where," Presented at the QIP 2026 poster session, 2025.
3. Z. Deng, B. Ghaemmaghami, A. K. Singh, B. Cho, **L. Orshansky**, M. Erez, and M. Orshansky, "Enhancing cross-category learning in recommendation systems with multi-layer embedding training," in *Proceedings of the 15th Asian Conference on Machine Learning, PMLR 222*, PMLR, 2023, pp. 263–278.
4. D. Doty, N. Kornerup, A. Luchsinger, **L. Orshansky**, D. Soloveichik, and D. Woods, "Harvesting brownian motion: Zero energy computational sampling," Appeared as the Best Student Poster at DNA29 Conference, 2023. arXiv: `2309.06957 [cs.DS]`.

Awards and Achievements

2024
- **NSF Graduate Research Fellowship**, awarded by the National Science Foundation for the funding of research pursuits in theoretical computer science. Approx. $150,000 value.
- **Dean's Honored Graduate**, College of Natural Sciences, UT Austin. This is the highest academic honor recognized by the college.
- **Honors Scholar with Research Distinction**, College of Natural Sciences, UT Austin.

2023
- **Outstanding Chinese Language Student**, Department of Asian Studies, UT Austin.

2022
- **Distinguished Honors Scholar (Top 4%)**, College of Natural Sciences, UT Austin.

2020
- **Rocco C. Caffarelli Scholarship**, awarded on academic merit by Frost Bank. $20,000 value.
- **Salisbury Endowed Turing Scholarship**, awarded on academic merit by University of Texas at Austin. $2,000 value.